are arranged in a highly organised crystal pattern which permits little movement. As the ice melts, the water molecules become disorganised and can move more freely. The movement of molecules becomes freer in the liquid phase and even more free in the vapour phase. In other words, we can say that the randomness of the water molecules increases, as ice melts into water or water evaporates. Both are spontaneous processes which result in a increase in randomness (entropy).

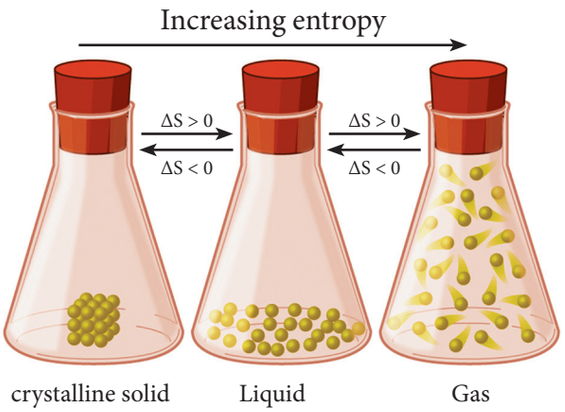


Figure 7.8 Illustration showing an increase in disorder.

Standard Entropy Change(ΔS^0):

It is possible to calculate the actual entropy of a substance at any temperature above 0 K. The absolute entropy of a substance at 298 K and one bar pressure is called the standard entropy S^o. The third law of thermodynamics states, according to Nernst, that the absolute entropy of elements is zero only at 0 K in a perfect crystal, and standard entropies of all substances at any temperature above 0 K always have positive values. Once we know the entropies of different substances, we can calculate the standard entropy change (ΔS_r^0) for chemical reactions.

$$\Delta S_r^0 = \sum S^0_{products} - \sum S^0_{reactants} \quad \text{------- (7.30)}$$

Standard Entropy of Formation:

Standard entropy of formation is defined as "the entropy of formation of 1 mole of a compound from the elements under standard conditions". It is denoted as ΔS_f^0. We can calculate the value of entropy of a given compound from the values of S^0 of elements.

Problem: 7.6

Calculate the standard entropy change for the following reaction(ΔS_f^0), given the standard entropies of $CO_2(g)$, $C(s)$,$O_2(g)$ as 213.6 , 5.740 and 205 JK^{-1} respectively.

$$C(g) + O_2(g) \longrightarrow CO_2(g)$$

$$\Delta S_r^0 = \sum S^0_{products} - \sum S^0_{reactants}$$

$$\Delta S_r^0 = \{S^0_{CO_2}\} - \{S^0_C + S^0_{O_2}\}$$

$$\Delta S_r^0 = 213.6 - [5.74 + 205]$$

$$\Delta S_r^0 = 213.6 - [210.74]$$

$$\Delta S_r^0 = 2.86\ JK^{-1}$$

Evaluate Yourself- 6

Urea on hydrolysis produces ammonia and carbon dioxide. The standard entropies of urea, H_2O, CO_2, NH_3 are 173.8, 70, 213.5 and 192.5J $mole^{-1}K^{-1}$ respectively. Calculate the entropy change for this reaction.

Entropy change accompanying change of phase

When there is a change of state from solid to liquid (melting), liquid to

11th Std Chemistry 186-228.indd 212 06-01-2021 15:46:19